

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Arithmetic Progressions

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

Find the 20th term of the AP 3, 7, 11, ...

Answer:

$$a_n = a + (n-1)d = 3 + 19(4) = 79.$$

Question 2

Find the 15th term of the AP 7, 12, 17, ...

Answer:

$$a_n = a + (n-1)d = 7 + 14(5) = 77.$$

Question 3

Find the 25th term of the AP 12, 15, 18, ...

Answer:

$$a_n = a + (n-1)d = 12 + 24(3) = 84.$$

Question 4

Find the 18th term of the AP 5, 11, 17, ...

Answer:

$$a_n = a + (n-1)d = 5 + 17(6) = 107.$$

Question 5

Find the 10th term of the AP 20, 18, 16, ...

Answer:

$$a_n = a + (n-1)d = 20 + 9(-2) = 2.$$

Question 6

Find the 12th term of the AP 9, 16, 23, ...

Answer:

$$a_n = a + (n-1)d = 9 + 11(7) = 86.$$

Question 7

Find the 14th term of the AP 2, 11, 20, ...

Answer:

$$a_n = a + (n-1)d = 2 + 13(9) = 119.$$

Question 8

Find the 16th term of the AP 15, 19, 23, ...

Answer:

$$a_n = a + (n-1)d = 15 + 15(4) = 75.$$

Question 9

Find the sum of the first 20 terms of the AP with $a = 2$ and $d = 3$.

Answer:

$$S_n = n/2[2a+(n-1)d] = 20/2[2(2)+19(3)] = 610.$$

Arithmetic Progressions - Continued

Question 10

Find the sum of the first 12 terms of the AP with $a = 5$ and $d = 5$.

Answer:

$$S_n = n/2[2a+(n-1)d] = 12/2[2(5)+11(5)] = 390.$$

Question 11

Find the sum of the first 15 terms of the AP with $a = 10$ and $d = 4$.

Answer:

$$S_n = n/2[2a+(n-1)d] = 15/2[2(10)+14(4)] = 570.$$

Question 12

Find the sum of the first 18 terms of the AP with $a = 3$ and $d = 6$.

Answer:

$$S_n = n/2[2a+(n-1)d] = 18/2[2(3)+17(6)] = 972.$$

Question 13

Find the sum of the first 10 terms of the AP with $a = 20$ and $d = -1$.

Answer:

$$S_n = n/2[2a+(n-1)d] = 10/2[2(20)+9(-1)] = 155.$$

Question 14

Find the sum of the first 11 terms of the AP with $a = 7$ and $d = 8$.

Answer:

$$S_n = n/2[2a+(n-1)d] = 11/2[2(7)+10(8)] = 517.$$

Question 15

Find the sum of the first 25 terms of the AP with $a = 12$ and $d = 2$.

Answer:

$$S_n = n/2[2a+(n-1)d] = 25/2[2(12)+24(2)] = 900.$$

Question 16

Find the sum of the first 16 terms of the AP with $a = 15$ and $d = 3$.

Answer:

$$S_n = n/2[2a+(n-1)d] = 16/2[2(15)+15(3)] = 600.$$

Question 17

Which term of the AP 4, 7, 10, ... is 34?

Answer:

$$4+(n-1)3=34; \text{ therefore } n-1=10 \text{ and } n=11.$$

Question 18

Which term of the AP 7, 12, 17, ... is 52?

Answer:

$$7+(n-1)5=52; \text{ therefore } n-1=9 \text{ and } n=10.$$

Question 19

Which term of the AP 2, 8, 14, ... is 56?

Answer:

$$2+(n-1)6=56; \text{ therefore } n-1=9 \text{ and } n=10.$$

Arithmetic Progressions - Continued

Question 20

Which term of the AP 10, 14, 18, ... is 46?

Answer:

$10 + (n-1)4 = 46$; therefore $n-1=9$ and $n=10$.

Question 21

State the main formula, rule or theorem used in Arithmetic Progressions.

Answer:

$a_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$.

Question 22

What do the important symbols or quantities in Arithmetic Progressions represent?

Answer:

a is first term, d is common difference, n is number of terms.

Question 23

Write the correct method for solving a Arithmetic Progressions numerical.

Answer:

Identify a , d and n , select a_n or S_n , then substitute carefully.

Question 24

Which unit, sign convention or condition must be checked in a Arithmetic Progressions answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Arithmetic Progressions be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Question 26

What is one common error in Arithmetic Progressions questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Question 27

How should full working be shown in a Arithmetic Progressions numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Arithmetic Progressions?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Arithmetic Progressions - Continued

Question 29

How can a student decide which formula to use in Arithmetic Progressions?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Arithmetic Progressions?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: [CBSE Mathematics curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); NCERT textbooks.

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